

How gamma controls access to a target during readout training

Small slopes can make an attainable target expensive to learn. In our frozen-feature tanh model, the same target reaches 1% relative error after **15.8 million GD updates at gamma 8**, compared with **16,013 updates at gamma 64**. Adam also struggles at gamma 8: error remains above 1% after 200,000 updates. Centers, target, readout coordinates, and zero initialization are held fixed.

The explanation has three parts. Gamma determines how strongly each feature smooths rapid variation. This changes the kernel's response to patterns in the target, and the measured target-weighted spectrum reveals their relative learning rates. Classical GD dynamics then predict the observed delay; Adam tests empirically how the difficulty persists under adaptive updates. We follow one representative target, then check other targets and optimizer settings.

Notation. Eigenvalues are denoted by μ throughout.

Symbol	Meaning
γ	Common slope of frozen tanh features
$J_\gamma, K_\gamma = J_\gamma J_\gamma^T$	Sample-normalized feature matrix and readout kernel
$y, r_n, e(n) = \ r_n\ /\ y\ $	Target samples, residual, and relative error
$\mu_i, u_i, \rho_i = \mu_i/\mu_1$	Eigenvalues, unit eigenvectors, relative eigenvalues
$p_i = u_i^T y ^2/\ y\ ^2$	Fraction of target energy in an eigenvector
$q_\gamma(v) = v^T K_\gamma v$	Kernel response to fixed unit sample pattern v

1 The target is attainable; the question is how quickly

The readout model is

$$\hat{f}(x) = w_0 + \sum_{j=1}^W w_j \tanh(\gamma(x - c_j)).$$

Only coefficients w_j are trained. Centers c_j and common slope γ remain fixed during each run. We compare $\gamma \in \{8, 12, 16, 64\}$ on the same uniform centers and samples. The primary target is

$$f(x) = \sin(2\pi x) + \frac{1}{2} \sin(6\pi x) + \frac{1}{4} \sin(10\pi x), \quad x \in [-1, 1].$$

There are 559 tanh features plus bias, spacing $h = 1/256$, and $m = 8,193$ training samples. Normalize features and target by $\sqrt{m}$:

$$J_\gamma[a, j] = \frac{\tanh(\gamma(x_a - c_j))}{\sqrt{m}}, \quad J_\gamma[a, 0] = \frac{1}{\sqrt{m}}, \quad y_a = \frac{f(x_a)}{\sqrt{m}}.$$

Then $\frac{1}{2}\|J_\gamma w - y\|^2$ is half mean-squared error. All four GD runs actually reach 1% error, establishing attainability at the studied accuracy, without requiring exact representation everywhere on the interval.

The kernel measures how residual patterns couple to available features. For a unit vector v on the sample grid,

$$q_\gamma(v) = v^T K_\gamma v = \|J_\gamma^T v\|^2.$$

This is total squared correlation of the dictionary with that pattern. For an eigenvector it equals the eigenvalue. For a general direction,

$$q_\gamma(v) = \sum_i \mu_i(\gamma) |u_i(\gamma)^T v|^2.$$

The same quadratic expression defines $q_\gamma(v)$ for arbitrary v ; unit normalization makes directional responses comparable. We can track a physically fixed pattern across gamma while allowing the eigenvectors to change.

2 Gamma explicitly controls the kernel response

Smaller gamma smooths more strongly

A tanh transition is a smoothed step:

$$\tanh(\gamma \cdot) = s_\gamma * \text{sign}, \quad s_\gamma(x) = \frac{\gamma}{2} \text{sech}^2(\gamma x).$$

Convolution averages translated steps against the density s_γ . Small gamma spreads this average more widely. Its Fourier multiplier is

$$M_\gamma(\omega) = \frac{z}{\sinh z}, \quad z = \frac{\pi|\omega|}{2\gamma}, \quad M_\gamma(0) = 1.$$

It is close to one for slow variation and decays rapidly when $|\omega|/\gamma$ is large. Increasing gamma admits more rapid variation. The squared multiplier enters the kernel because it pairs two smoothed features. This describes original tanh features; the step explains their construction.

An explicit statement for the finite tanh dictionary

Inputs and centers stay fixed. For $d_{ab} = x_a - x_b$, define the analytic center-integral contribution and finite correction:

$$B_\gamma[a, b] = \frac{W+1}{m} - \frac{2}{hm} d_{ab} \coth(\gamma d_{ab}), \quad R_\gamma = K_\gamma - B_\gamma.$$

Use the continuous value $d \coth(\gamma d) = 1/\gamma$ at zero. R_γ accounts for the finite center interval and replacement of its integral by a discrete sum. These effects belong to the actual model. The proof gives their explicit integral expression; the example measures their size.

Theorem 1 (Gamma-dependent response of a finite tanh kernel). For consecutive uniformly spaced centers, fixed input samples, slopes $\Gamma > \gamma > 0$, and every real sample vector v , define

$$V_v(\omega) = \sum_{a=1}^m v_a e^{-i\omega x_a}.$$

Then

$$q_\Gamma(v) - q_\gamma(v) = \mathcal{A}_{\Gamma,\gamma}(v) + v^T (R_\Gamma - R_\gamma) v, \\ \mathcal{A}_{\Gamma,\gamma}(v) = \frac{2}{\pi h m} \int_{\mathbb{R}} \frac{M_\Gamma(\omega)^2 - M_\gamma(\omega)^2}{\omega^2} |V_v(\omega)|^2 d\omega \geq 0.$$

The apparent zero-frequency singularity has a continuous limit. No zero-mean assumption on v is needed.

The multiplier difference is the explicit gamma gain; $|V_v|^2$ describes the tested pattern's frequencies; $1/\omega^2$ comes from underlying step features. The last term accounts for finite geometry. Its sign and size depend on direction, so a positive Fourier contribution alone does not prove monotonicity of every finite-kernel eigenvalue or ratio.

Follow one direction the target actually needs

At gamma 8, select the eigenvector with largest target energy among resolved positive modes satisfying $\rho_i \leq 2 \times 10^{-6}$. This reproducible retrospective selection uses the initial target and kernel, with no training outcomes. It selects **eigenvector 22**, carrying **4.14% of target squared norm**, with relative eigenvalue only 1.612×10^{-6} .

Freeze this vector v and change gamma. Starting from its gamma-8 response, the explicit mechanism predicts

$$q_{\Gamma}^{\text{Fourier}}(v) = q_8(v) + \mathcal{A}_{\Gamma,8}(v).$$

This uses the known reference direction and gamma multiplier, with no new eigenspace calculation at other slopes. Compare it with the directly measured $\|J_{\Gamma}^T v\|^2$. Numerically, the Fourier term is evaluated through its equivalent closed-form pairwise entries $2[d \coth(8d) - d \coth(\Gamma d)]/(hm)$, proved below. Response rises from 0.000389636 at gamma 8 to 0.386738 at gamma 64, about a thousandfold. At gammas 12, 16, and 64, finite corrections are respectively **0.0601%**, **0.0339%**, and **0.00439%** of actual response. On this target-relevant direction, the explicit Fourier contribution quantitatively explains the change.

The direction stays fixed. At larger gamma, its response is a weighted average of new eigenvalues, not necessarily one eigenvalue. Small measured corrections apply to this probe and geometry. The full measured spectrum below shows how change is distributed among actual learning directions.

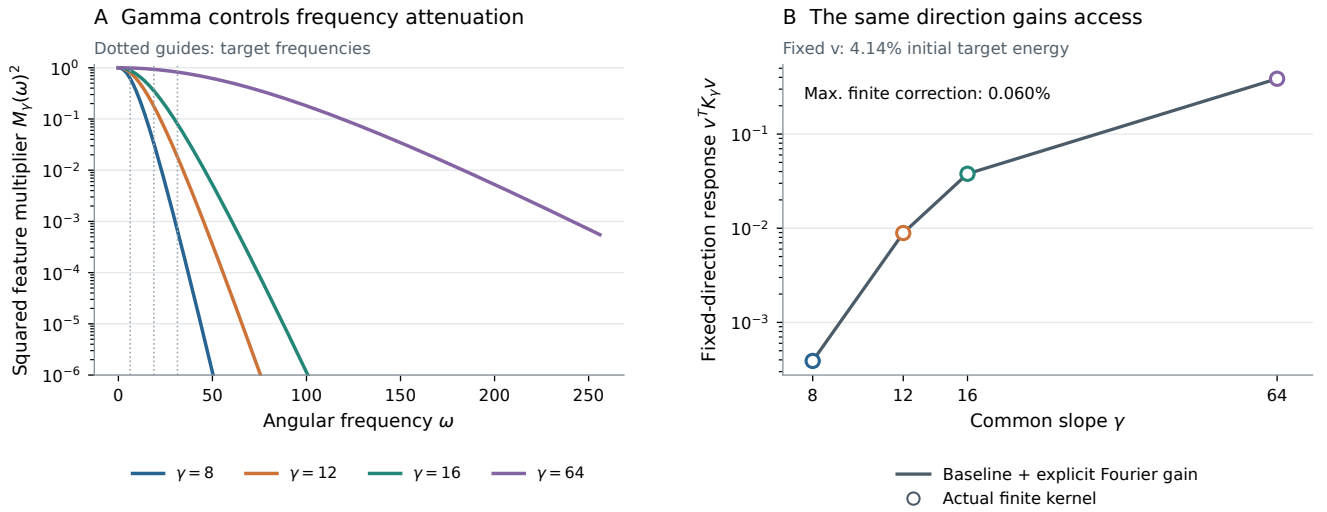


Figure 1: Gamma’s action is explicit and measurable. Left: analytic squared multipliers at all four gammas; marked frequencies are the target’s three nominal frequencies, not automatically finite-kernel eigenmodes. Right: actual response and reference response plus Fourier gain for fixed gamma-8 eigenvector 22, carrying 4.14% initial target energy. Curves nearly coincide because finite correction is below 0.061% of actual response at every larger gamma. These are directional responses, not eigenvalues tracked across gamma.

3 The measured spectrum explains the GD delay

For zero initialization, $r_n = J_\gamma w_n - y$ obeys

$$r_{n+1} = (I - \eta K_\gamma) r_n.$$

Expanding y in orthonormal eigenvectors gives each component the factor $1 - \eta \mu_i$. Orthogonality yields exact relative squared error:

$$e(n)^2 = \sum_i p_i(\gamma) (1 - \eta \mu_i(\gamma))^{2n}.$$

Nullspace energy is included with factor one. The largest eigenvalue limits the shared stable step: $0 < \eta < 2/\mu_1$. Our runs use $\eta \mu_1 \approx 1/2$, giving, at the exact normalized step,

$$e(n)^2 = \sum_i p_i(\gamma) \left(1 - \frac{\rho_i(\gamma)}{2}\right)^{2n}.$$

Forecasts use the actual archived step. A ratio of one halves a component's residual each update; a ratio of 10^{-6} takes about two million updates for one e-fold reduction. Target weights determine which rates matter. Large eigenvalues elsewhere cannot accelerate an independently evolving slow GD component.

For the fixed direction above, measured response divided by $\mu_1(\gamma)$ is 1.612×10^{-6} at gamma 8 and 1.560×10^{-3} at gamma 64, about a 968-fold increase. This is the direction's weighted average of relative eigenvalues. The complete target-weighted spectrum accounts for how the individual rates contribute to training.

We measure the finite tanh spectrum and target weights, then evaluate this classical formula without fitting a rate to training trajectories. First 1% crossings are **15,798,313; 186,057; 61,792; and 16,013 updates** at gammas 8, 12, 16, and 64. All four agree with executed crossings. The gamma mechanism explains the kernel-response change; the spectral calculation accounts for complete finite geometry when computing time.

4 Does Adam encounter the same difficult directions?

Adam rescales readout-coordinate updates and uses gradient history:

$$r_{n+1} = r_n - \alpha_n J_\gamma D_n \hat{m}_n, \quad D_n = \text{diag}\left((\sqrt{\hat{v}_n} + \epsilon_{\text{Adam}})^{-1}\right).$$

$\hat{m}_n$ and $\hat{v}_n$ are bias-corrected first and second moments. Even without momentum, the effective kernel changes with D_n . GD eigenvectors are diagnostic coordinates for Adam; GD decay is not an Adam time law.

We can still ask: **does error remain in directions the raw kernel learns slowly?** Project saved Adam residuals onto the finite-kernel modes:

$$\sum_{0 < \rho_i(\gamma) \leq \rho} \frac{|u_i(\gamma)^T r_n|^2}{\|y\|^2}.$$

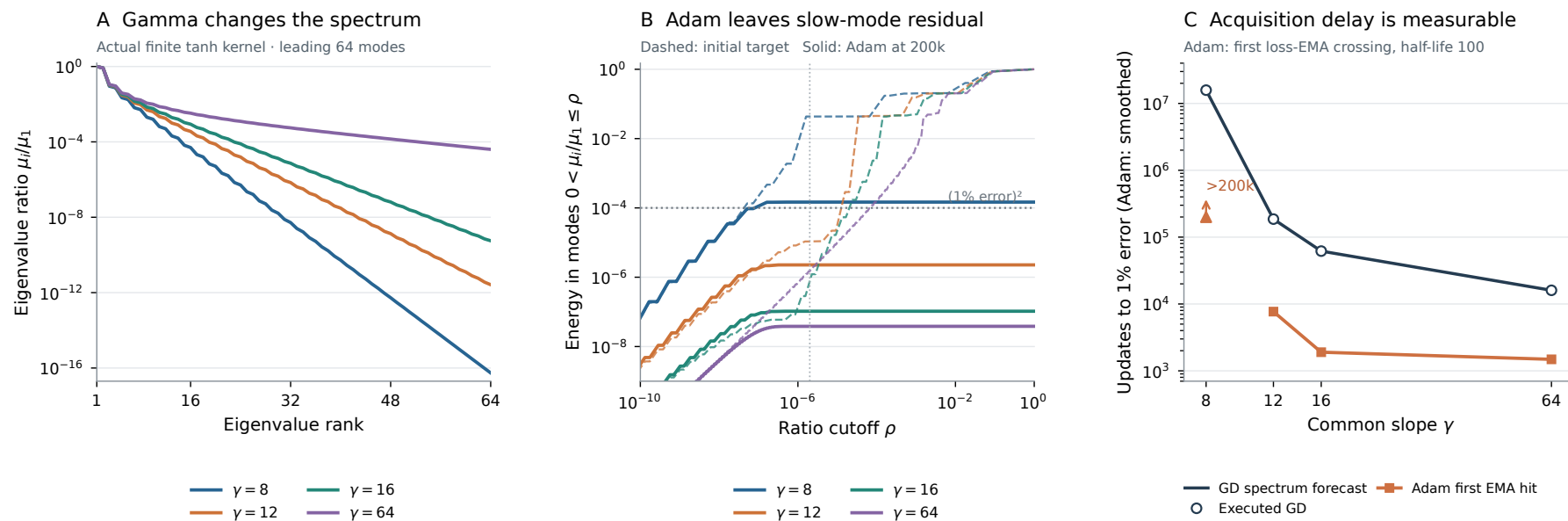


Figure 2: From gamma to learning difficulty. A: leading 64 actual finite-kernel eigenvalue ratios, ordered separately at each gamma; matching rank does not imply matching eigenvectors. B: initial target energy and residual energy after 200,000 common-recipe Adam updates, accumulated over positive modes below each ratio cutoff. Energy 10^{-4} is the 1% error budget. C: executed GD crossings and measured-spectrum forecasts, with observed first Adam loss-EMA crossings using one 100-update half-life. The arrow means no crossing by 200,000; GD has longer budgets. Target and feature geometry match across panels.

All energies use original target norm. At gamma 8, modes with $\rho_i \leq 2 \times 10^{-6}$ initially hold 4.33% of target energy. After 200,000 Adam updates, remaining energy is 1.47094×10^{-4} , alone above the 10^{-4} budget for 1% error. Almost all final residual energy lies there. Adam removes about 99.66% of initial band energy, but enough remains to prevent acquisition.

At gammas 12, 16, and 64, final band energies are 2.2660×10^{-6} , 1.0437×10^{-7} , and 3.8307×10^{-8} , below tolerance. Subspaces share a spectral definition across gamma, while eigenvectors may differ. Adam can temporarily increase projected residual components; these measurements describe remaining error, not untouched initial content.

Make the Adam acquisition statistic visible

The common setting uses initial rate 10^{-3} , epsilon 10^{-12} , and moments (0.9, 0.999). Rate stays constant through 20,000 updates, decays by cosine to 10^{-6} at 50,000, then remains fixed through 200,000. This shared schedule permits an early-learning comparison while explaining later settling behavior.

Smooth squared relative error with shared half-life $s = 100$ updates:

$$M_0 = e(0)^2 = 1, \quad M_n = \beta M_{n-1} + (1 - \beta)e(n)^2, \quad \beta = 2^{-1/s}.$$

The first update with $M_n \leq 10^{-4}$ is first EMA acquisition. Plots show $\sqrt{M_n}$ in relative-error units. Common-recipe crossings are **7,688**; **1,900**; and **1,486** for gammas 12, 16, and 64; gamma 8 never crosses within the horizon. Every crossing is computed from every saved update.

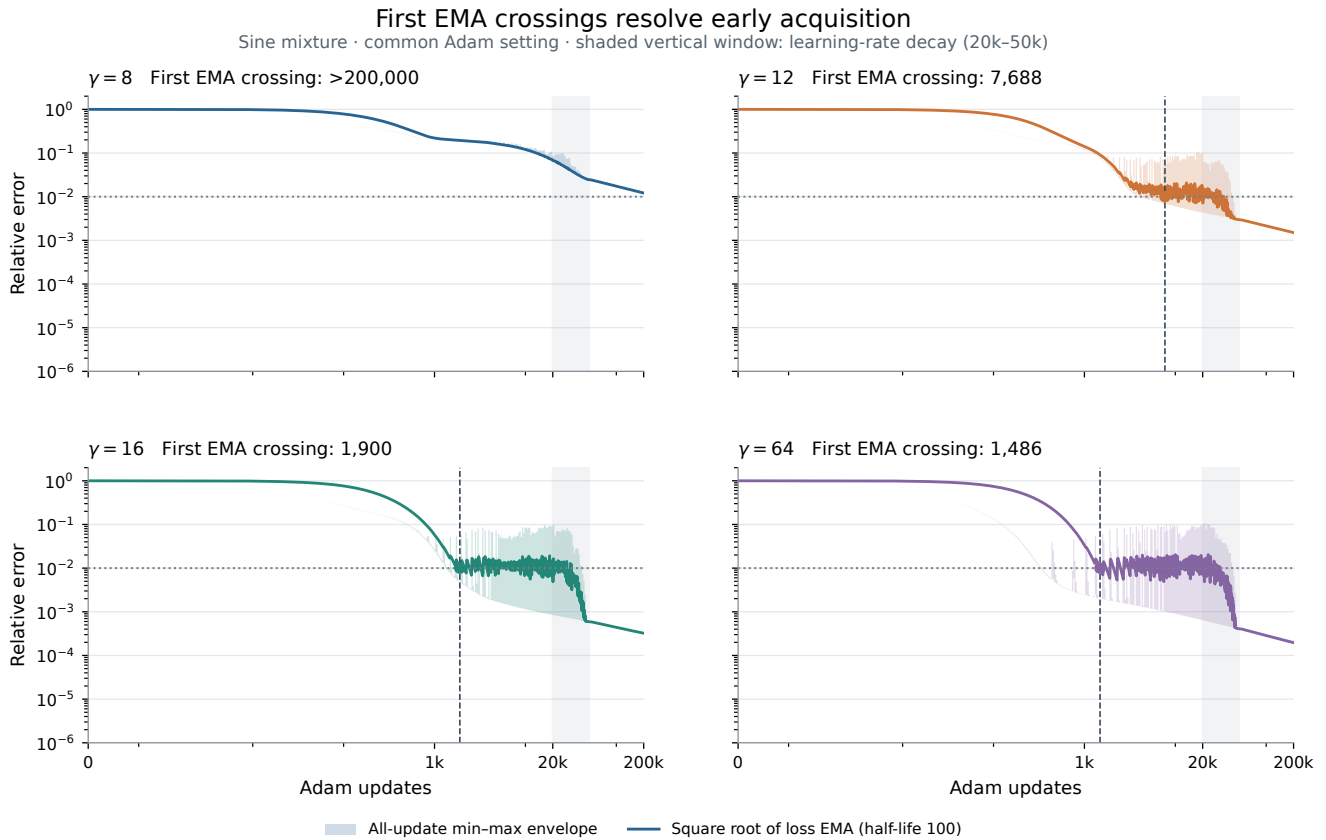


Figure 3: The crossing comes from actual training. Bands show minima and maxima of raw errors in update bins; lines show the square root of loss EMA. Dashed vertical lines and threshold dots identify first crossings. The gray window is shared rate decay from 20,000 to 50,000. Displayed curves are sampled; crossing calculations use every update.

This measures first acquisition of smoothed loss. Later EMA upcrossings occur 73, 97, and 98 times at the three successful slopes. Raw sustained crossings occur around 41,000 under the shared schedule. The EMA window is a retrospective visualization choice with a memory-induced resolution limit: $M_n \geq \beta^n$ precludes crossing before $\lceil s \log_2(10^4) \rceil$. At half-life 100 this floor is **1,329 updates**, close to the gamma-64 crossing. EMA compresses differences between sufficiently fast runs.

For this attainable target and fixed geometry, small gamma leaves needed directions weakly coupled to features. GD exhibits the delay implied by its spectrum; Adam retains sufficient error in those slow directions to miss the same accuracy threshold at gamma 8 within its budget.

5 Proof of the gamma-response theorem

Let $I = [c_1 - h/2, c_W + h/2]$ be the union of center cells, and put

$$F_{ab,\gamma}(c) = \tanh(\gamma(x_a - c)) \tanh(\gamma(x_b - c)).$$

First, the whole-line deficit is

$$\int_{\mathbb{R}} (1 - F_{ab,\gamma}(c)) dc = 2d_{ab} \coth(\gamma d_{ab}).$$

For distinct samples, use $1 - \tanh u \tanh v = \coth(u - v)(\tanh u - \tanh v)$. The translated-profile difference integrates to twice its translation. For coincident samples, $\int \operatorname{sech}^2(\gamma c) dc = 2/\gamma$ gives the continuous limit.

Since $|I| = Wh$, subtract the deficit and restore its part outside I to obtain $K_\gamma = B_\gamma + R_\gamma$, with

$$\begin{aligned} R_\gamma[a, b] &= \frac{1}{m} \left[\sum_{j=1}^W F_{ab,\gamma}(c_j) - \frac{1}{h} \int_I F_{ab,\gamma}(c) dc \right] \\ &\quad + \frac{1}{hm} \int_{\mathbb{R} \setminus I} (1 - F_{ab,\gamma}(c)) dc. \end{aligned}$$

The bracket is the quadrature discrepancy; the last integral accounts for the finite interval. Bias contributes $1/m$ to B_γ and cancels between gammas.

Next, $s_\gamma * \operatorname{sign} = \tanh(\gamma \cdot)$ follows by differentiation: both derivatives equal $2s_\gamma$ and both functions vanish at zero. With $\widehat{g}(\omega) = \int g(x) e^{-i\omega x} dx$, substitute $u = e^{2\gamma x}$ and $a = \omega/(2\gamma)$:

$$\begin{aligned} \widehat{s}_\gamma(\omega) &= \int_0^\infty \frac{u^{-ia}}{(1+u)^2} du = \Gamma(1-ia)\Gamma(1+ia) \\ &= \frac{\pi a}{\sinh(\pi a)} = M_\gamma(\omega). \end{aligned}$$

These equalities use the beta integral and gamma-function reflection identity. Here $\Gamma(\cdot)$ is the gamma function, distinct from the larger slope Γ .

Define the integrable difference profile

$$D(d) = 2[d \coth(\gamma d) - d \coth(\Gamma d)].$$

Differentiating the deficit integral twice and integrating once by parts gives

$$\frac{d^2}{dd^2} [2d \coth(\gamma d)] = 4(s_\gamma * s_\gamma)(d).$$

To see every step, write the deficit as $\int [1 - t_\gamma(u)t_\gamma(u - d)] du$, with $t_\gamma(u) = \tanh(\gamma u)$. Its second derivative is

$$-\int t_\gamma(u)t_\gamma''(u - d) du = \int t_\gamma'(u)t_\gamma'(u - d) du.$$

The boundary term vanishes, and $t_\gamma' = 2s_\gamma$, proving the identity. Consequently

$$\widehat{D}(\omega) = \frac{4(M_\Gamma(\omega)^2 - M_\gamma(\omega)^2)}{\omega^2}, \quad \widehat{D}(0) = \frac{\pi^2}{3}(\gamma^{-2} - \Gamma^{-2}).$$

This follows from $\widehat{D}'' = -\omega^2 \widehat{D}$; expand $z/\sinh z$ to get the continuous zero-frequency value. Since $z/\sinh z$ decreases for positive z , $M_\Gamma \geq M_\gamma$, so the transform is nonnegative.

Finally $B_\Gamma[a, b] - B_\gamma[a, b] = D(x_a - x_b)/(hm)$. Fourier inversion contributes $1/(2\pi)$, and summation against $v_a v_b$ yields $|V_v(\omega)|^2$. This proves the Fourier term with prefactor $2/(\pi hm)$. Add the exact finite correction to prove the theorem. $\square$

6 Empirical checks and reproduction

The five targets are the sine mixture, $\exp(\sin(3\pi x))$, $1/(1 + 25x^2)$, $\sqrt{5}x^2$, and $\sqrt{2}\sin(2\pi x)$. Supplementary comparisons retain shared and archived validation-selected Adam settings. Selection used median validation error at five checkpoints from 40,000 to 50,000, over five initial rates and two epsilons; optimizer state continued intact. These plots measure training-grid optimization, without a new held-out generalization claim.

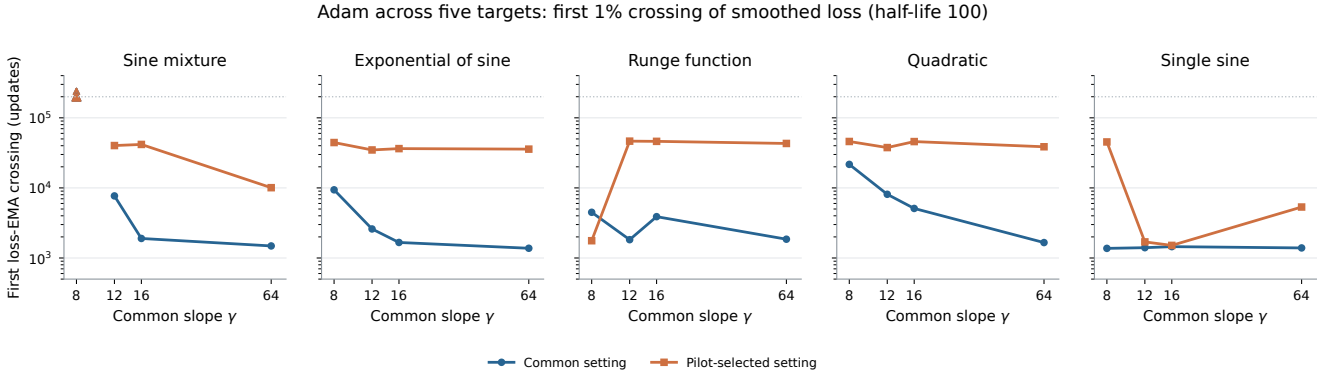


Figure 4: Target and optimizer setting matter. All archived targets use the same 100-update loss-EMA half-life. Selected settings optimize late validation error, not early acquisition. Arrows indicate no crossing within 200,000. Results do not yield universal monotone ordering by gamma.

The primary common-recipe ordering appears at EMA half-lives 30, 100, and 300. At 1,000, crossings cluster near 28,000–30,000 and reorder. With selected settings, half-life-100 crossings at gammas 12, 16, and 64 are 40,208; 41,716; and 10,076. These expose dependence on the measurement window and optimizer recipe.

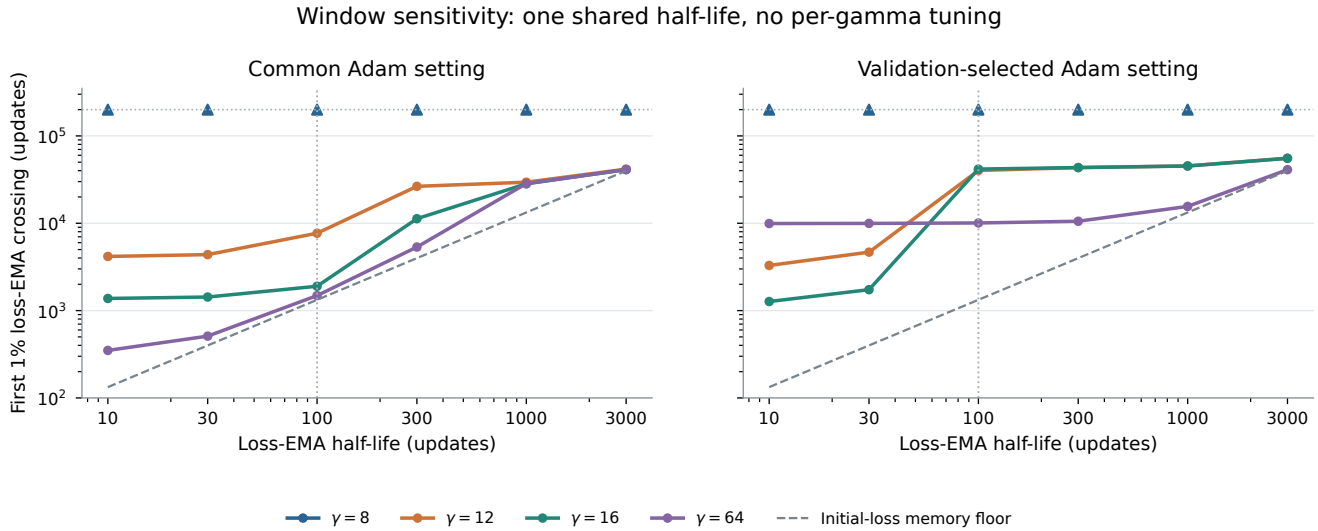


Figure 5: The smoothing choice is visible. Each horizontal position uses one half-life for all gammas. The dotted line marks 100 updates; the gray dashed line is earliest possible crossing due to initial-loss memory. Triangles mark censoring at 200,000. All points summarize the same saved trajectories.

All calculations use archived FP64 features and trajectories. Modal analysis retains singular values above 10^{-14} of the largest; omitted numerical directions are reported separately, not identified as exact nullspace. Independent SVD and direct residual projections reproduce slow-band energies within 5.1×10^{-14} . Every-update traces determine EMA crossings; saved readouts determine modal residual measurements. The fixed-direction calculation compares the explicit increment to direct feature action and records the signed finite correction. No new training was performed.

The [numerical evidence](#) contains the selected direction, plotted quantities, source hashes, and verification records. Reproduce from the repository root with one BLAS thread:

```
export OPENBLAS_NUM_THREADS=1 VECLIB_MAXIMUM_THREADS=1
python -m experiments.expD36_frozen_gamma_probe.gamma_mechanism_analysis
python -m experiments.expD36_frozen_gamma_probe.adam_spectral_analysis
python -m experiments.expD36_frozen_gamma_probe.adam_ema_analysis \
    --archive /path/to/adam_raw_scalar_archive.npz
python -m experiments.expD36_frozen_gamma_probe.optimizer_access_figure
latexmk -pdf -outdir=/tmp/gamma-optimizer-latex \
    docs/gamma_optimizer_access_note.tex
```

Extract the scalar archive from persistent training data using `adam_trace_audit.py`. Existing compact EMA artifacts suffice to regenerate figures without retrieving complete traces.